

# Day 03 Hybrid Notes

Transformations on graph paper and Desmos | Cloze + outline + active learning

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|-------|-------|---------|
| Name: | Date: | Period: |
|-------|-------|---------|

**Learning Targets**  
I can describe translations, reflections, rotations, and dilations.  
I can transform points on the coordinate plane.  
I can tell which transformations preserve distance and angle measure.

**Part 1: Cloze Notes - Essential Ideas**  
1. A translation slides a figure without turning or \_\_\_\_\_.  
2. A reflection flips a figure across a \_\_\_\_\_.  
3. A rotation turns a figure around a \_\_\_\_\_.  
4. A dilation changes size by a \_\_\_\_\_.  
5. Rigid motions preserve distance and angle \_\_\_\_\_.  
6. A dilation with scale factor greater than 1 is an \_\_\_\_\_.  
7. A dilation with scale factor between 0 and 1 is a \_\_\_\_\_.

|                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                          |
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| <b>Part 2: Outline Notes - Big Picture</b><br>A. Transformations describe how a preimage moves to an _____.<br>B. Translations, reflections, and rotations are _____.<br>C. Dilations keep shape but may change _____.<br>D. Coordinate rules make transformations _____. | <b>Connection to Prior Math</b><br>1. What algebra skill shows up today?<br>_____<br>2. What diagram or graph skill shows up today?<br>_____<br>3. What is one place this could appear outside math class?<br>_____<br>— |
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**Most Important Reminder**  
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

# Day 03 Hybrid Notes

Guided setup + active learning

## Part 3: Guided Setup Practice

1. Translate  $P(-2, 5)$  by  $(x + 4, y - 3)$ .

Setup/work: \_\_\_\_\_

Answer: \_\_\_\_\_

2. Rotate  $A(3, -1)$  90 degrees counterclockwise about the origin.

Setup/work: \_\_\_\_\_

Answer: \_\_\_\_\_

## Part 4: Error Analysis

A student reflects  $(4, -7)$  over the x-axis and gets  $(-4, -7)$ .  
What went wrong?

\_\_\_\_\_

Correct idea:

\_\_\_\_\_

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## Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

## Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

## Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

\_\_\_\_\_.

Sentence 2: One thing I need to remember when solving is

\_\_\_\_\_.

## Part 5C: Application Problem

A logo point  $(2, 6)$  is dilated by scale factor 3 from the origin. What is the image?

Setup: \_\_\_\_\_

Answer: \_\_\_\_\_

Sentence answer: \_\_\_\_\_